RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Thursday, 24 September 2026, 8:22 AM
State	Finished
Completed on	Thursday, 24 September 2026, 8:23 AM
Time taken	1 min
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int countOccurrences(int a[], int low, int high, int candidate)
{
    int count = 0;
    for (int i = low; i <= high; i++)
    {
        if (a[i] == candidate)
            count++;
    }

    return count;
}
int majorityElement(int a[], int low, int high)
{
    if (low == high)
        return a[low];
    int mid = (low + high) / 2;
    int leftMajority = majorityElement(a, low, mid);
    int rightMajority = majorityElement(a, mid + 1, high);

    if (leftMajority == rightMajority)
        return leftMajority;

    int leftCount = countOccurrences(a, low, high, leftMajority);
    int rightCount = countOccurrences(a, low, high, rightMajority);

    if (leftCount > rightCount)
        return leftMajority;
    else
        return rightMajority;
}

int main()
{
```

```
35      int n;
36      scanf("%d", &n);
37
38      int a[n];
39
40      for (int i = 0; i < n; i++)
41      {
42          scanf("%d", &a[i]);
43      }
44
45      int result = majorityElement(a, 0, n - 1);
46
47      printf("%d", result);
48
49      return 0;
50  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course